Service Bulletin

Service Bulletin Number: 5504538	Released Date: 11-jul-2018
Fuel Filter Plugging	

Fuel Filter Plugging

Product Affected

- All High Horsepower (HHP) Engines

Symptoms

- Fuel filter life is **not** meeting the published maintenance interval

How To Use This Tree

This symptom tree is to be used to troubleshoot reduced fuel filter life. Perform the list of troubleshooting steps in the sequence shown in the Specifications/Repair section of the tree.

Shoptalk

Fuel filters protect the fuel system by significantly reducing the number of hard particles present in fuel. Fuel restriction will increase over time as sediment gets collected in the filter media. When fuel filter restriction increases and fuel filter life does not meet expectations, the source of the contaminant can be understood and reduced or eliminated.

Common causes of reduced fuel filter life:

1. Hard particle (ISO 4406) contaminants such as silicon and aluminum can be attributed to fuel cleanliness and handling. Modern fuel systems operate with component match clearances typically from 2 to 5 microns for injectors. At these pressures, very small, hard particles are potential sources of fuel system malfunction. Engine builders and fuel injection equipment manufacturers have found that the particles of size approaching the 4 and 6 micron ISO checkpoints are particularly critical to the durability of the fuel injection system.
2. Water enters diesel fuel at various locations along the supply chain and becomes a serious issue when present as free water. Water contributes to corrosion, biological contamination, and injector deposits.
3. Microbial contaminants such as bacteria, yeast, and mold grow into a large colony of microorganisms in the presence of water. These colonies cause gel like substances to form on filter media and block fuel flow. Indications of microbial growth include:
    a. Slime deposits on tank walls, piping, or other surfaces which are exposed to fuel. These deposits are usually greenish-black or brown and are slick to the touch.
    b. Black or brown "stringy" material suspended in tank water bottoms.
    c. Swelling or blistering of any rubber surface (washers, hoses, connectors, and so forth) that comes in contact with fuel.
    d. Sludge or slime deposits on filter surfaces.

e. Foul odor resembling that of rotten eggs (hydrogen sulfide).

4. Asphaltenes form as fuel is recirculated and exposed to high pressures and temperatures during continuous operation. Asphaltenes begin to form larger clusters of insoluble materials that cause diesel fuel to appear black in color and cause high filter restriction.
5. Oil mixing with fuel can cause the diesel fuel to appear black in color. The presence of oil in fuel could be intentional from the practice of blending lubricating oil with fuel to be burned by the engine such as the Centinel oil replenishment system or could occur due to engine oil mixing with the fuel during operation.
6. Some cold flow improvers can plug the high efficiency filter media.
7. Additive presence and concentration is not always known and can lead to chemical interactions that cause gel like substances to form on filter media and block fuel flow.
8. Fuel wax occurs naturally in diesel fuel. Solubility of the wax is dependent on temperature. As fuel cools, a temperature is reached at which the soluble paraffin wax in the fuel begins to come out of solution (Cloud Point); any further cooling will cause wax to separate out of solution. The temperature at which a certain fuel will become saturated with wax and causes filter plugging problems is termed the Cold Filter Plugging Point (ASTM D6371).

Overview of the troubleshooting process listed in this tree:

- Fuel can become contaminated at various points in the delivery chain including after the fuel has been delivered to the customer. This procedure utilizes the Fleetguard® Portable Fuel Cleanliness Analysis Kit (part number FK36000) available from Cummins Filtration to identify where in the system the fuel is contaminated. For example fuel may not pass the fuel cleanliness criteria upon delivery, at a bulk storage tank, at onsite fuel delivery trucks, and/or at the equipment fuel tank.
- Results of the Fleetguard® Portable Fuel Cleanliness Analysis Kit will allow specific recommendations for using fuel test kits or sending fuel samples or the fuel patch for further analysis to identify the source of contaminant.
- After identifying the contaminant, corresponding actions are recommended to immediately address the concern and prevention reoccurrence of issue.

Troubleshooting Summary

STEPS		**SPECIFICATIONS**
STEP 1.	Identify where in the system the fuel is contaminated using the Fleetguard® Portable Fuel Cleanliness Analysis Kit. If only one piece of equipment out of an entire fleet is plugging fuel filters, start at step 1C.	
	STEP 1A. Test fuel as delivered to site from the fuel delivery truck	Fuel quality is clean?

STEPS		SPECIFICATIONS
	STEP 1B. Test fuel from onsite fuel delivery trucks and bulk tanks, etc (if applicable)	Fuel quality is clean?
	STEP 1C. Test fuel from equipment fuel tank at the inlet of the stage 1 fuel filter	Fuel quality is clean?
STEP 2.	Bulk fuel tank and onsite fuel delivery troubleshooting procedures.	
	STEP 2A. Test fuel for bacteria, mold, and yeast	Fuel tests positive for bacteria, mold, or yeast?
	STEP 2B. Test fuel for hard particles	Fuel within specification?
	STEP 2C. Test fuel for water contamination	Fuel within specification?
	STEP 2D. Test fuel patch for chemical contamination	Chemical contamination detected?
STEP 3.	Equipment fuel tank troubleshooting procedures.	
	STEP 3A. Test fuel for bacteria, mold, and yeast	Fuel tests positive for bacteria, mold, or yeast?
	STEP 3B. Test fuel for hard particles	Fuel within specification?
	STEP 3C. Test fuel for water contamination	Fuel within specification?
	STEP 3D. Test fuel patch for chemical contamination	Chemical contamination detected?
	STEP 3E. Test fuel for required fuel properties	Fuel within specification?
	STEP 3F. Visually observe the color of the fuel.	Is fuel black?

STEP 1. Identify where in the system the fuel is contaminated using the Fleetguard® Portable Fuel Cleanliness Analysis Kit. If only one piece of equipment out of an entire fleet is plugging fuel filters, start at step 1C.

STEP 1A. Test fuel as delivered to site from the fuel delivery truck

Conditions : **None**		
Action	**Specification/Repair**	**Next Step**
Use Fleetguard® Portable Fuel Cleanliness Analysis Kit (part number FK36000) to determine if the fuel is good upon delivery.	Fuel quality is clean according to FK36000 instruction? **YES**	**1B**
	Fuel quality is clean? **NO** **Repair :** Reject fuel and work with fuel supplier to improve quality.	**1B**

STEP 1B. Test fuel from onsite fuel delivery trucks and bulk tanks, etc (if applicable)

Conditions: **None**		
Action	**Specification/Repair**	**Next Step**
Collect fuel after fuel tank bulk filters, fuel delivery truck filters, etc. Use Fleetguard® Portable Fuel Cleanliness Analysis Kit (part number FK36000) to determine if the fuel is good.	Fuel quality is clean according to FK36000 instruction? **YES**	**1C**
	Fuel quality is clean? **NO** **Repair :** Retain patch in a clean bag or container and label with location where fuel was collected.	**2A**

STEP 1C. Test fuel from equipment fuel tank at the inlet of the stage 1 fuel filter.

Conditions: **Equipment fuel tank must be ½ full or less**		
Action	**Specification/Repair**	**Next Step**
Collect fuel at inlet of stage 1 fuel filter. Use Fleetguard® Portable Fuel Cleanliness Analysis Kit (part number FK36000) to determine if the fuel is good.	Fuel quality is clean according to FK36000 instruction? **YES**	**Repair complete**
	Fuel quality is clean? **NO** **Repair :** Retain patch in a clean bag or container and label with location where fuel was collected.	**3A**

STEP 2. Bulk fuel tank and onsite fuel delivery troubleshooting procedures.

STEP 2A. Test fuel for bacteria, mold, and yeast.

Conditions: **None**		
Action	**Specification/Repair**	**Next Step**
Use fuel tank sampling kit CC2523 to collect a fuel sample from the bottom of the fuel tank. Sample must be taken from the bottom of the tank where water accumulates. Test fuel for bacteria, mold, and yeast using the test strips in kit CC2524.	Bacteria, mold or yeast present? **YES** **Repair :** Drain and clean the fuel tank To prevent reoccurrence of bacteria, mold, and yeast: Regularly drain the fuel tank of all free water, consider implementing kidney loop filtration	**2B**
	Bacteria, mold or yeast present? **NO**	**2B**

STEP 2B. Test fuel for hard particles.

Conditions: **None**		
Action	**Specification/Repair**	**Next Step**
Collect a fuel sample at the outlet of the fuel storage tank or fuel delivery truck after bulk tank or fuel delivery filtration. Submit fuel sample for particle count (ISO 4406). Reference Fleetguard Monitor^{TM} part number CC2719.	Fuel meets ISO Code 18/16/13? **YES**	**2C**
	Fuel meets ISO Code 18/16/13? **NO** **Repair :** Verify presence and condition of inspection lids and seals, visually inspect for presence of dipstick covers, verify presence and maintenance of fuel tank breathers or dessicant breathers, visually inspect for presence and maintenance of fuel delivery system filters, visually inspect for presence of supply and delivery fitting covers, visually inspect for method of hose storage and whether hose ends are capped, consider implementing kidney loop filtration or increasing fuel delivery system filtration as needed	**2C**

STEP 2C. Test fuel for water contamination.

Conditions:
None

Action	**Specification/Repair**	**Next Step**
Collect a fuel sample at the outlet of the fuel storage tank or fuel delivery truck after bulk tank or fuel delivery filtration. Submit fuel sample for water contamination test ASTM D6304 ISO 12937. Reference Fleetguard Monitor™ part number CC2719.	Fuel meets spec of less than 200 ppm water? **YES**	**2D**
	Fuel meet spec of less than 200 ppm water? **NO** **Repair :** Drain the fuel tank of all free water, verify presence and condition of inspection lids and seals, visually inspect for presence of dipstick covers, verify presence and maintenance of fuel tank breathers or dessicant breathers, visually inspect for presence and maintenance of fuel delivery system filters, visually inspect for presence of supply and delivery fitting covers, visually inspect for method of hose storage and whether hose ends are capped, consider implementing kidney loop filtration or increasing fuel delivery system filtration as needed	**2D**

STEP 2D. Test fuel patch for chemical contamination.

Conditions: **None**		
Action	**Specification/Repair**	**Next Step**
Send patch from step **1B** to laboratory for SEM- low mag and high mag, EDAX- elemental composition, FTIR- list the functional groups and match against a library Reference Cummins Filtration Website https://www.cumminsfiltration.com/fluidanalysis for further instructions.	Chemical contaminants present? **YES** **Repair :** Work with laboratory for recommendations on how to minimize chemical contaminants	**Repair Complete**
	Chemical contaminants present? **NO**	**Repair Complete**

STEP 3. Equipment fuel tank troubleshooting procedures.

STEP 3A. Test fuel for bacteria, mold, and yeast.

Conditions: **Equipment fuel tank must be ½ full or less**		
Action	**Specification/Repair**	**Next Step**
Use fuel tank sampling kit CC2523 to collect a fuel sample from the bottom of the fuel tank. Sample must be taken from the bottom of the tank where water accumulates. Test fuel for bacteria, mold, and yeast using the test strips in kit CC2524.	Bacteria, mold or yeast present? **YES** **Repair :** Drain and clean the fuel tank To prevent reoccurrence of bacteria, mold, and yeast: Regularly drain the fuel tank of all free water	**3B**
	Bacteria, mold or yeast present? **NO**	**3B**

STEP 3B. Test fuel for hard particles.

Conditions: **Equipment fuel tank must be ½ full or less**		
Action	**Specification/Repair**	**Next Step**
Collect a fuel sample at the inlet of the stage 1 fuel filter. Submit fuel sample for particle count (ISO 4406). Reference Fleetguard Monitor™ part number CC2719.	Fuel meets ISO Code 18/16/13? **YES**	**3 C**
	Fuel meets ISO Code 18/16/13? **NO** **Repair :** Verify presence of fast fill dry break system to reduce contamination during refueling, verify presence and maintenance of a 3 micron fuel tank vent filter, drain and clean the fuel tank or circulate fuel from tank through external kidney loop filtration	**3 C**

STEP 3C. Test fuel for water contamination.

Conditions: **Equipment fuel tank must be ½ full or less**		
Action	**Specification/Repair**	**Next Step**
Collect a fuel sample at the inlet of the stage 1 fuel filter. Submit fuel sample for water contamination test ASTM D6304 ISO 12937. Reference Fleetguard Monitor™ part number CC2719.	Fuel meets spec of less than 200 ppm water? **YES**	**3D**
	Fuel meets spec of less than 200 ppm water? **NO** **Repair :** Drain the fuel tank of all free water, verify presence and maintenance of a 3 micron fuel tank vent filter, drain and clean the fuel tank or circulate fuel from tank through external kidney loop filtration	**3D**

STEP 3D. Test fuel patch for chemical contamination.

Conditions: **Equipment fuel tank must be ½ full or less**		
Action	**Specification/Repair**	**Next Step**
Send patch from step **1 C** to laboratory for SEM- low mag and high mag, EDAX- elemental composition, FTIR- list the functional groups and match against a library Reference Cummins Filtration Website https://www.cumminsfiltration.com/fluidanalysis for further instructions.	Chemical contaminants present? **YES** **Repair :** Work with laboratory for recommendations on how to minimize chemical contaminants.	**3E**
	Chemical contaminants present? **NO**	**3E**

STEP 3E. Test fuel for required fuel properties.

Conditions: **Equipment fuel tank must be ½ full or less**		
Action	**Specification/Repair**	**Next Step**
Collect a fuel sample at the inlet of the stage 1 fuel filter. Submit fuel sample for fuel quality test. Reference Fleetguard Monitor™ part number CC2651.	Fuel meets fuel property requirements in table 1 of the Fuels for Cummins Engines service bulletin, 3379001? **YES**	**3F**
	Fuel meets property requirements in table 1 of the Fuels for Cummins Engines service bulletin, 3379001? **NO** **Repair :** Reject fuel and work with fuel supplier to meet required fuel properties.	**3F**

STEP 3F. Visually observe the color of the fuel.

<table>
<tr><td colspan="3">Conditions:
Equipment fuel tank must be ½ full or less</td></tr>
<tr><th>Action</th><th>Specification/Repair</th><th>Next Step</th></tr>
<tr><td rowspan="2">Collect a fuel sample at the inlet of the stage 1 fuel filter. Visually observe the color of the fuel.</td><td>Is fuel black?
YES
Repair :
Engines using Centinel oil replenishing systems which discard used oil into the fuel return tank will appear black. Inspect the Centinel oil control valve for proper operation. Refer to the Centinel Oil Control Valve procedure 007-078 in the Centinel Master Repair Manual, bulletin 3666231.
If Centinel is not being used, and fuel is black, check for sources of oil contamination into fuel system. Submit fuel sample for asphaltene test method ASTM D6560. If asphaltenes are present, use an asphaltene conditioner.</td><td>Repair complete</td></tr>
<tr><td>Is fuel black?
NO</td><td>Repair complete</td></tr>
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Document History

Date	Details
2018-7-11	Module Created